

Projects of DataBase System

Lecturer: Shuai Wang

TA: Hao Gong

Projects Outline

0 - Project Preparation

1 - Project One

2 - Project Two

3 - Project Three

4 - Project Four

5 - Project Five

6 - Project Six

Every 2 Week

0 - Project Preparation

- I. Project Rules
- II. Software Preparation
- III. Some Common DBMS
- IV. Frontend Technology
- V. Project One Preview

Project Rules

【Grading Criteria】

- ✓ **Mid Exam + Lab Work + Others : 30%**
- ✓ **Final Exam: 70%**

Project Rules

【DeadLine】

Before 23:59 of the next Project.

Project Rules

【Grace Day Policy】

- **If you have a Grace Day Policy, you can ask for a project to delay a day's acceptance.**
- **You can only use Grace Day Policy once in a project.**
- **Earn them from the class.**

Project Rules

【Submitt Assignment】

- Send the lab report and experimental results required by the project to the email address “seudatabaseqa@gmail.com” with the title “**StudentNumber-Name-Project-X**” and the lab report will be used as the basis for the rating.
- For Example: “71117315-凌泰炜-Project-1”

0-Project Preparation

I. Project Rules

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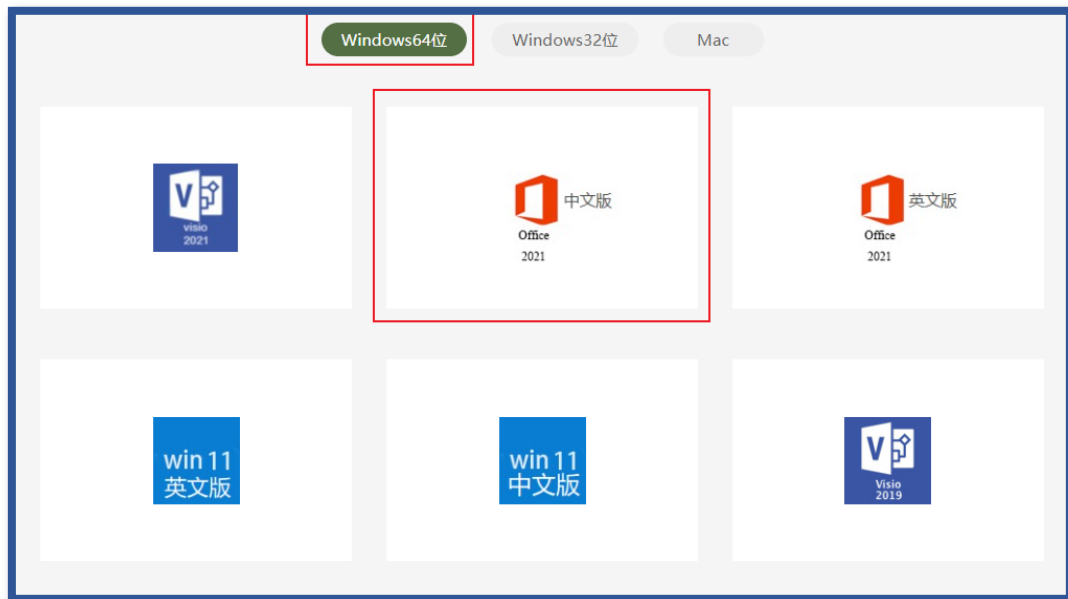
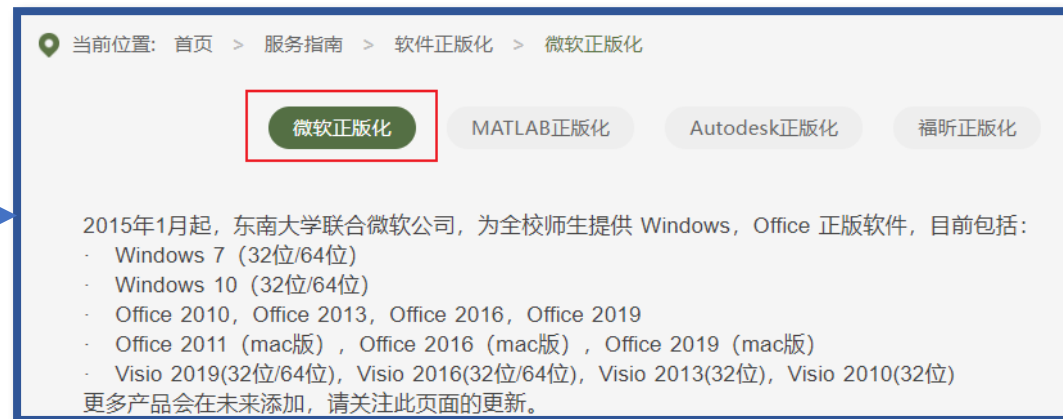
About Access

- A **database management system (DBMS)** from **Microsoft**
- Combines the **relational Access Database Engine (ACE)** with a **graphical user interface** and **software-development tools**.
- A member of the **Microsoft Office** and **Microsoft 365** suite of application.

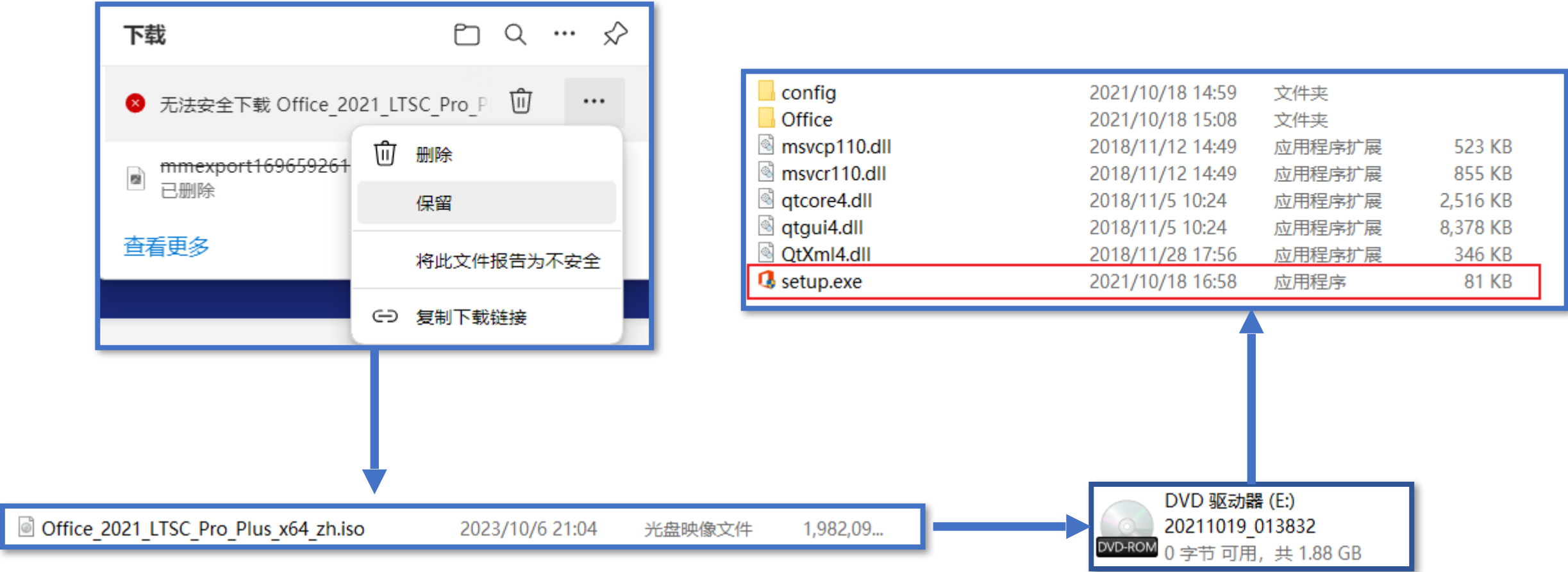


Download MS Access from SEU

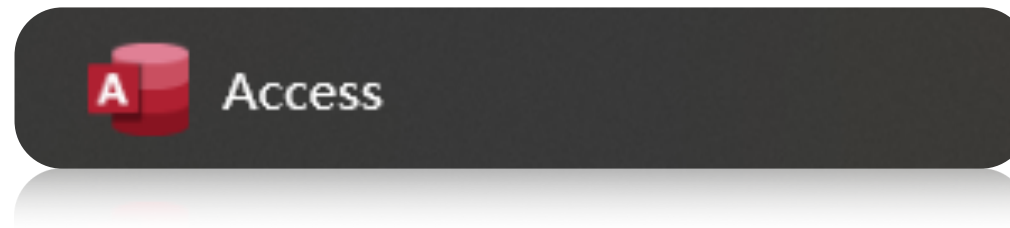
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激活解决方案-东南大学网络与信息中心 (seu.edu.cn)

软件安装许可协议

本站提供软件仅供东南大学在校师生员工下载、安装使用，禁止向校外任何个人和组织分发相关软件的软件副本，违者将承担相应的法律责任。

☐ 我已阅读并同意以上协议

微软正版软件下载

微软激活解决方案

1、KMS激活

适用于校园网内用户，使用方式简单，只需在待激活的电脑上，以管理员权限运行激活脚本，并选择要激活的项目，即可完成激活。首次激活成功后，系统会定期（周期为180天）连接校内KMS服务器自动完成激活。

 [激活脚本\(解压后请用管理员身份运行\).zip](#)

产品信息

Office

产品已激活

Microsoft Office LTSC 专业增强版 2021

本产品包含



[更改产品密钥](#)

- 1.Office 2010 激活
- 2.Office 2013 激活
- 3.Office 2016 激活
- 4.Office 2019/2021 激活
- 5.Windows 激活

请输入数字并回车:4

0-Project Preparation

- I. Project Rules
- II. Software Preparation
- III. Some Common DBMS
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1. MySQL

- Latest version 8.0. The language used is **C and C++**.
Relational database. Works on many different platforms : Linux / Windows /Solaris...
- Few features of this tool are:
 - (1)High-speed data processing;
 - (2)Use of **triggers** increases productivity, with **rollback** and **commit** helps in data recovery if required.
- Costs: Open-source and free for Community Server.
Commercial fees for Enterprise Edition
- Website: [MySQL](https://www.mysql.com/)

“LAMP”



2. SQLite

- The latest release version is 3.39.3. Coded in **C language**.
It is mainly used as a database system for mobiles. It can work on Linux, Windows, and Mac operating systems.
- Few features of this tool are:
 - (1) Concise functionality, small size.
 - (2) Don't need much space
 - (3) Fast and don't need to set up.
- Costs: It's an open-source tool.
- Website: [SQLite](https://www.sqlite.org/)



3. Redis

- The latest stable release is 7.0.4.

NoSQL, work on Windows and Linux operating systems. It is coded in ANSI C language.

- Few features of this tool are:

(1) supports **distributed systems**.

(2) **key-value** storage system

(3) **high processing speed**. data types like **hashes** and **strings** are also supported.

- Costs: It's an open-source tool.

- Website: [Redis](https://redis.io)



4. Neo4j

- The latest stable version is 5.12. Coded in **Java**
Graph database(NoSQL), used on Windows and Linux/Unix operating systems.
- Few features of this tool are:
 - (1)Stores data in the form of **graphs (nodes, edges, and attributes of nodes and edges)**.
 - (2)Using **Cypher query language (CQL)**, easy to learn.
 - (3) easy to represent and manage data in **connected relationships**.
- Costs: Open-source and free for Community Server.
Commercial fees for Enterprise Edition
- Website: [Neo4j](https://neo4j.com)



5. OrientDB

- The latest stable version is 3.2.23. It is coded in **Java** language **NoSQL** , used on Windows and Linux platforms.
- Few features of this tool are:
 - (1) A **graphical database**.
 - (2) Widely used across big data market and in real-time web-based applications.
- Costs: It's an open-source tool.
- Website: [OrientDB](http://OrientDB.com)



6. phpMyAdmin

- The latest stable release is 5.2.1. It is coded primarily in **PHP**. It can work on Windows and Linux operating systems.
- Few features of this tool are:
 - (1) Use the **web interface** to manage **MySQL databases**.
 - (2) **Interface** is user-friendly, data can be exported in **CSV, SQL, XML** files and it can be imported from both **CSV** and **SQL** file formats.
- Costs: It's an open-source tool.
- Website: [phpMyAdmin](https://www.phpmyadmin.net/)



7. SQL Developer

- The latest stable release is 23.1. It is coded in **Java**.
It can work on Windows and Linux operating systems.
- Few features of this tool are:
 - (1) **Graphical interface.**
 - (2) Less execution time.
 - (3) Queries can be run and generated in **many formats** such as HTML, PDF, XML, and Excel.
- Costs: It's an open-source tool.
- Website: [SQL Developer](https://www.oracle.com/technetwork/database/developer-tools/sql-developer/index-089476.html)



0-Project Preparation

I. Project Rules

II. Software Preparation

III. Some Common DBMS

IV. Frontend Technology

V. Project One Preview

About Project Six

- Unlike Project One-Five, Project Six is a **TEAM** project
- In Project Six, you need to use Frontend Technology

Join GitHub

Create your account

Username *

Email address *

Password *

Make sure it's at least 15 characters OR at least 8 characters including a number and a lowercase letter. [Learn more.](#)

Email preferences

☒ Send me occasional product updates, announcements, and offers.

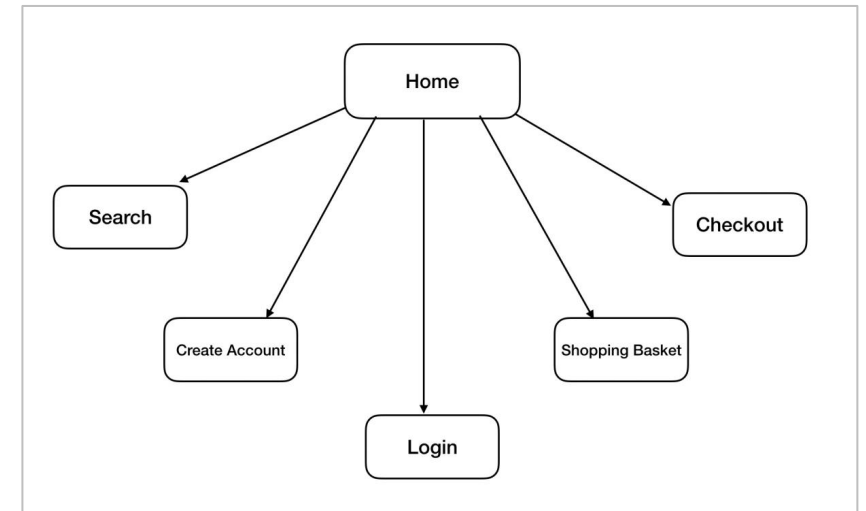


Sign in to GitHub

Username or email address

Password [Forgot password?](#)

New to GitHub? [Create an account.](#)



About Project Six

✓ “Internet Bookstore”

- B/S or C/S
- Database: Access , MySQL , ...



Frontend Technology

- HTML

HTML provides the basic structure of sites, which is enhanced and modified by other technologies like CSS and JavaScript.

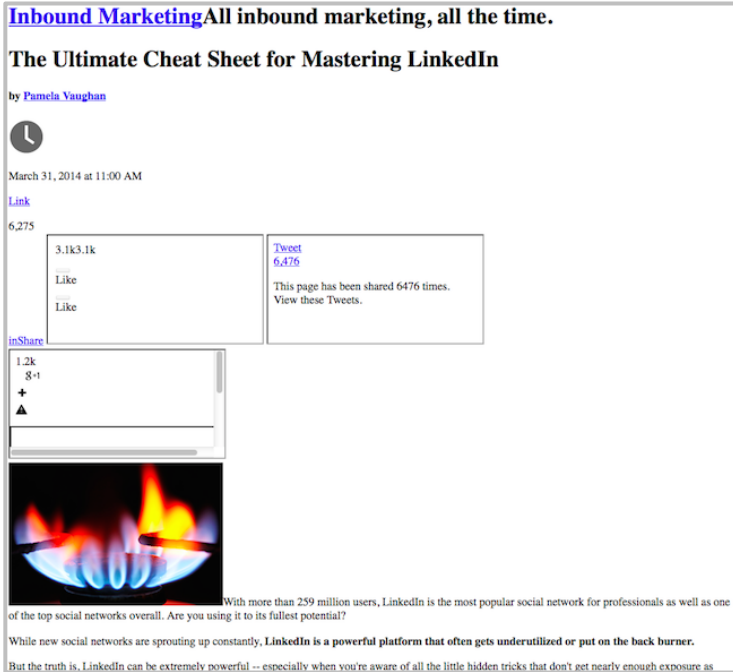
- CSS

CSS is used to control presentation, formatting, and layout.

- JavaScript

JavaScript is used to control the behavior of different elements.

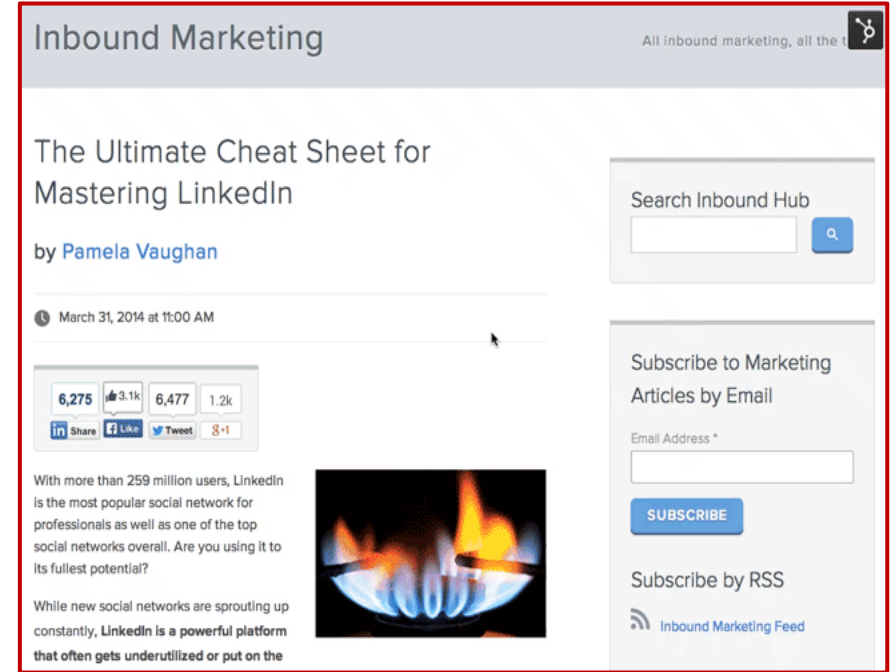
Frontend Technology



Only HTML



Enhanced by
CSS



Enhanced by
CSS and JavaScript

Framework Tools

■ React

React's reusable components ensure that developers don't have to rewrite the same code over and over again.



Framework Tools

■ Vue

Excellent documentation and intended for beginners and seasoned developers alike; An extensive tooling ecosystem that has grown over the years.



Framework Tools

■ Angular

Extremely complex web apps can be developed on the enterprise scale that rival desktop-based applications.



Development Tools

- Frontend

WebStorm, Visual Studio Code, Adobe Dreamweaver

- Backend

IDEA

- Collaboration

GitHub

0-Project Preparation

- I. Introduce Myself and Rules
- II. Software Preparation
- III. Some Common DBMS
- IV. Frontend Technology
- V. Project One Preview

Project One Preview

- Create a Database
- Create a Table
- Set the Field Properties for an Existing Table
- Add Data to a Table
- Import Data into Access

Project 1

Lecturer: Shuai Wang

TA: Hao Gong

Requirements for submitting the assignment

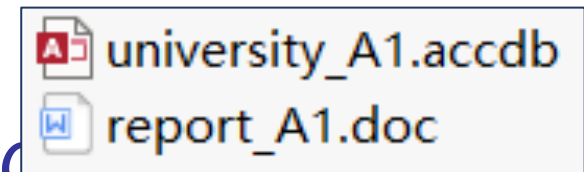
- **Email address :** seudatabaseqa@gmail.com
- **File name format :** StudentNumber-Name-Project-X

example : *71121000-张三-Project-1.zip*

- **Including :**

report_A1.doc / report_A1.docx

university_A1.mdb / university_A1.accdb



Deadline: Beijing time, 10.18, 24:00:00

Project 1

I. Tutorial with Project Examples

II. Assignment

Tutorial with Project Examples

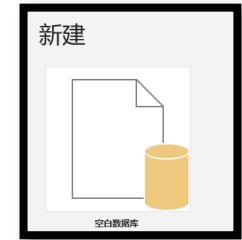
1. Create a Database

[Microsoft Access 2016 Tutorial \(quackit.com\)](http://quackit.com)

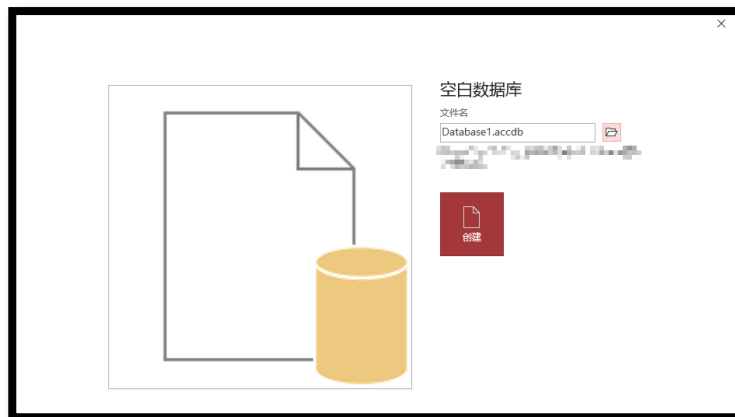
1.1 Select a Blank Database

1.1.1 From the File > Open > Choose One Database

1.1.2 From the File > New Folder > Blank Database



1.2 Name the Database



Tutorial with Project Examples

2. Create a Table

2.1 Customize the Blank Table

2.1.1 Select a Data Type for the New Field

2.1.2 Name the Field

2.1.3 Save the Table

2.1.4 Name the Table

Field Name	Data Type
AlbumId	Auto-Coding
AlbumName	Short Text
ReleaseDate	Date & Time
ArtistId	Number
GenreId	Number

AlbumId ▾	AlbumName ▾	ReleaseDate ▾	ArtistId ▾	GenreId ▾
1	JAY	2000/11/7	1	1
2	Fantasy	2001/9/20	1	1
3	The Eighth Dimension	2002/7/18	1	1
4	Seven Colored Fragrance	2004/8/3	1	1
5	Chopin's November	2005/11/1	1	1

Tutorial with Project Examples

2. Create a Table

2.2 Create a Table in Design View

2.2.1 Create the Table

2.2.2 Switch to Design View

2.2.3 Add Fields in Design View

2.2.4 Set a Primary Key

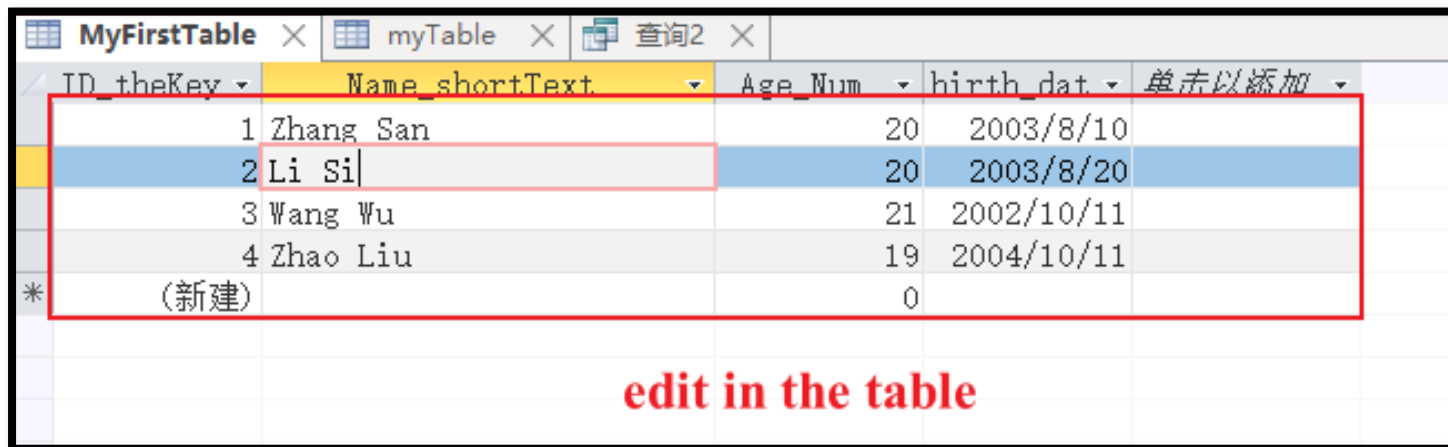
2.2.5 Save the Table

	字段名称	数据类型
表1	AlbumId	自动编号
	AlbumName	短文本
	ReleaseDate	日期/时间
	ArtistId	数字
	GenreId	数字

Tutorial with Project Examples

3. Add Data to Table

3.1 Add Data Directly in Datasheet View



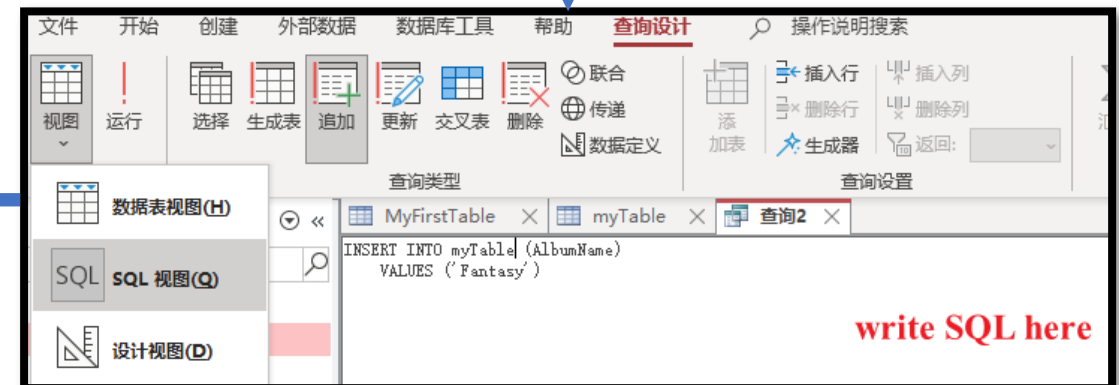
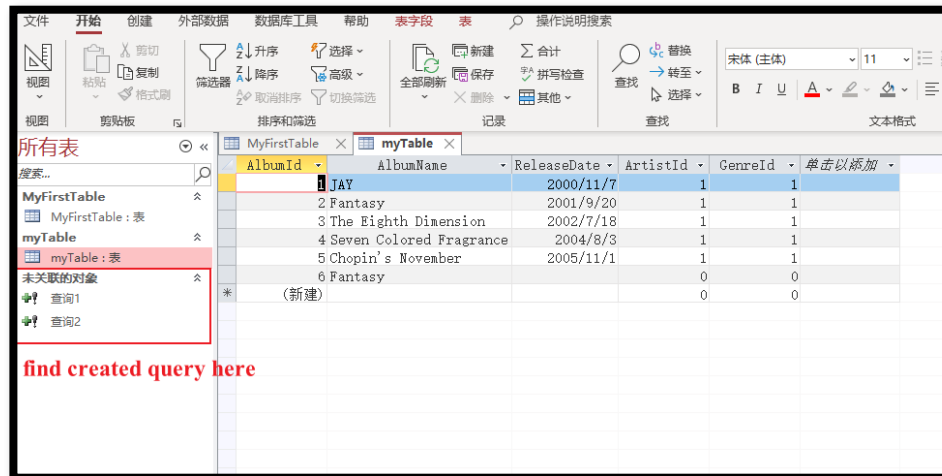
ID_theKey	Name_shortText	Age_Num	hirth_dat	单击以添加
1	Zhang San	20	2003/8/10	
2	Li Si	20	2003/8/20	
3	Wang Wu	21	2002/10/11	
4	Zhao Liu	19	2004/10/11	
*	(新建)	0		

edit in the table

Tutorial with Project Examples

3. Add Data to Table

3.2 Add Data using SQL

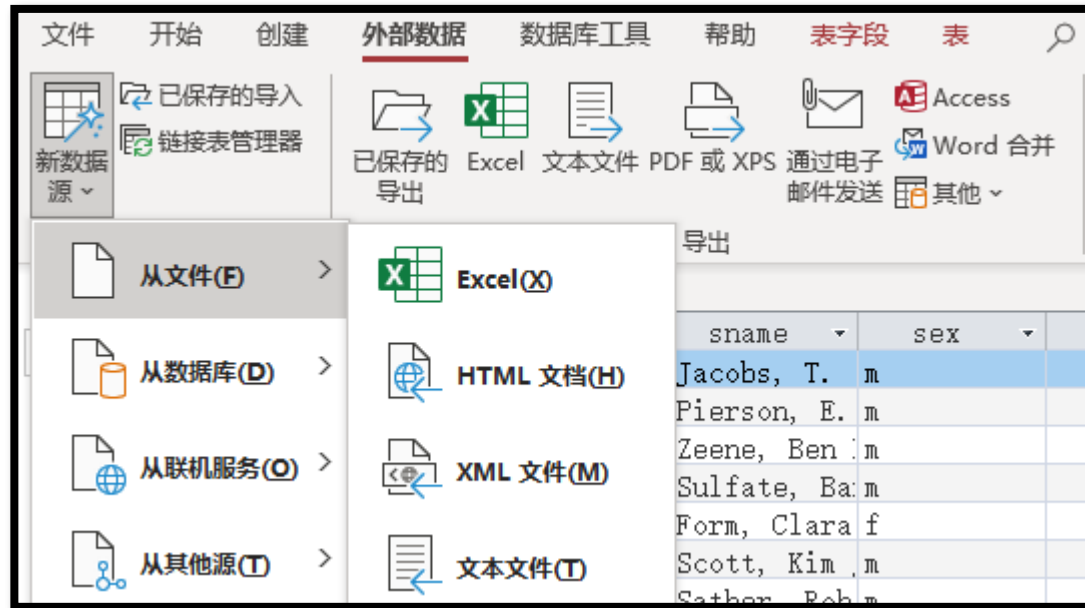


Little Hard

Tutorial with Project Examples

3. Add Data to Table

3.3 Add Data from External Resource



Follow the guides of Access to complete step by step.

Project 1

I. Tutorial with Project Examples

II. Assignment

Assignment

1. Assignment Content

1.1 Familiar with MS Access

1.2 Create a university database

1.2.1 university.mdb

1.2.2 Load data into the database

1.2.3 Use SQL to insert one record

1.3 Open the university database

1.3.1 Use C++/Python to generate more than 5000 random tuples for *student* table

1.3.2 Use SQL to insert those random tuples into database

Assignment

2. Schema and Sample Data

- The schema of the database is provided below (keys are red, field types are omitted):
 - student(sid, sname, sex, age, year, gpa)
 - dept(dname, numphds)
 - prof(pname, dname)
 - course(cno, cname, dname)
 - major(dname, sid)
 - section(dname, cno, sectno, pname)
 - enroll(sid, grade, dname, cno, sectno)

Multiple keys

	字段名称	数据类型
key	sid	数字
	grade	数字
	dname	短文本
key	cno	数字
key	sectno	数字

Assignment

2. Schema and Sample Data

- Before you start writing SQL, it is a good idea to take a look at the database and familiarize yourself with its contents.
- Write down as many details of your experiment as possible, and combining **TEXT** and **PICTURE** will get better grades.

Assignment

3. Data Files

- Download the following data files from WeChat Group
 - course.txt
 - dept.txt
 - enroll.txt
 - major.txt
 - prof.txt
 - section.txt
 - student.txt